HOSPITAL OPERATIONS DATA ANALYSIS

Patient Visits, Waiting Time and Service Performance

PROJECT TYPE	Exploratory Data Analysis (EDA)
DATASET	500 synthetic hospital visit records
TOOLS	Python, Pandas, NumPy, Matplotlib, Jupyter Notebook
SUBMITTED BY	_____
COURSE / CLASS	_____
SUBMITTED TO	_____

Academic project | Educational synthetic data only

1. Executive Summary

This project analyses **500** synthetic hospital visit records to understand patient flow and service performance. It focuses on department-wise visits, waiting time, consultation duration, billing, admissions and satisfaction.

The overall average waiting time is **18.8 minutes**. Average patient satisfaction is **4.44 out of 5**, and the average recorded bill is approximately ■**7,240**.

A total of **68** visits (13.6%) are labelled as admitted in this simulated dataset. These values are descriptive outputs from generated data, not real hospital performance indicators.

2. Introduction

Hospitals handle different types of patient visits across multiple departments. Understanding waiting time, service duration, admission patterns and patient satisfaction can help management identify areas for operational improvement.

Data analysis converts raw records into useful summaries and visualizations. This project uses a small, structured dataset and a beginner-friendly Python workflow to demonstrate that process.

3. Problem Statement and Objectives

Problem statement: A fictional hospital wants a simple overview of patient visits and service performance but has not yet organized its information into meaningful summaries.

Objectives: (1) inspect and clean the data; (2) measure department-wise patient volume; (3) study waiting and consultation times; (4) compare billing and admissions; (5) review satisfaction scores; and (6) present findings through charts.

4. Project Prompt

You are a junior data analyst working with a fictional hospital. Use the supplied dataset to explore how patient visits differ by department and visit type. Check data quality, clean missing numeric values, calculate summary statistics, create six charts and write a short conclusion based on the results.

5. Dataset Description

The dataset contains 500 synthetic records and 14 columns. Each row represents one fictional hospital visit. Patient_ID is an artificial record code; no real patient identity or medical history is included.

Column	Description
Patient_ID	Artificial visit record identifier
Age / Gender	Basic demographic fields
Department	Hospital department visited
Visit_Type	OPD or Emergency
Wait_Time_Minutes	Waiting time before consultation
Consultation_Minutes	Consultation duration
Length_of_Stay_Hours	Recorded stay duration
Billing_Amount_INR	Simulated bill amount in rupees
Insurance	Whether insurance is recorded
Admission_Status	Admitted or Not Admitted
Satisfaction_Score	Rating from 1 to 5
Follow_Up_Required	Yes or No

Column	Description
Visit_Month	Month of visit

6. Tools and Technologies

Python: programming language used to perform the analysis.

Pandas: used to load, clean, group and summarize tabular data.

NumPy: used for numerical operations and synthetic data generation.

Matplotlib: used to create charts and visualizations.

Jupyter Notebook: used to organize code, outputs and explanations.

7. Methodology

Step 1 – Data loading: read the CSV file into a Pandas DataFrame.

Step 2 – Data inspection: review dimensions, column types, missing values and duplicate rows.

Step 3 – Data cleaning: fill missing waiting time, satisfaction and billing values with each column's median.

Step 4 – Analysis: calculate descriptive statistics and group averages by department or visit type.

Step 5 – Visualization: create six charts to communicate the main patterns.

Step 6 – Interpretation: summarize patterns while keeping the synthetic nature and limitations clear.

8. Data Cleaning and Preparation

The original file contains 18 missing cells across selected numeric columns. Median imputation is used for waiting time, satisfaction score and billing amount because it is a simple method that is less affected by extreme values than the mean.

After cleaning, the analysis table contains 500 rows and 14 columns. The Patient_ID field is used as an identifier, not as a numeric measurement. The dataset is checked for duplicate rows before analysis.

9. Descriptive Statistics

Average waiting time: **18.8 minutes**.

Average consultation duration: **18.0 minutes**.

Average length of stay: **6.4 hours**.

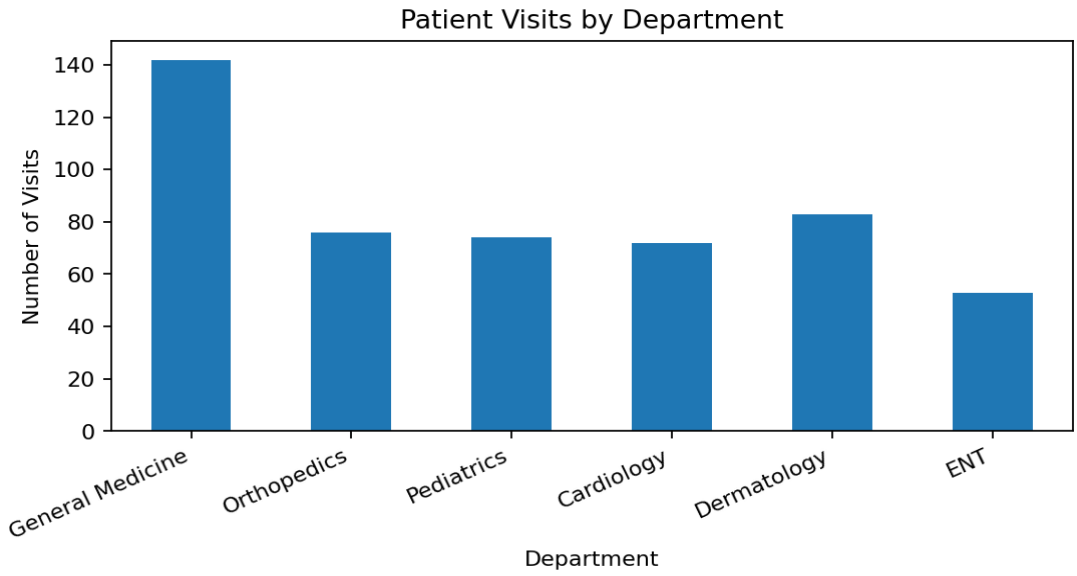
Average simulated bill: ■**7,240**.

Average satisfaction: **4.44/5**.

10. Results and Visualizations

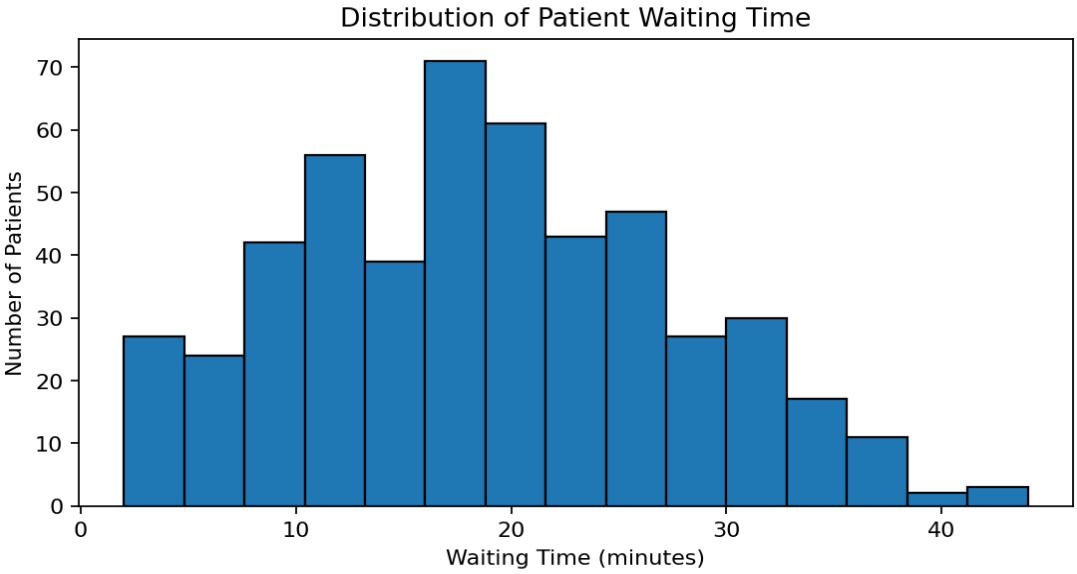
The following charts are generated from the cleaned dataset. Each chart answers a different operational question. Interpretations refer only to the synthetic records.

10.1 Patient Visits by Department



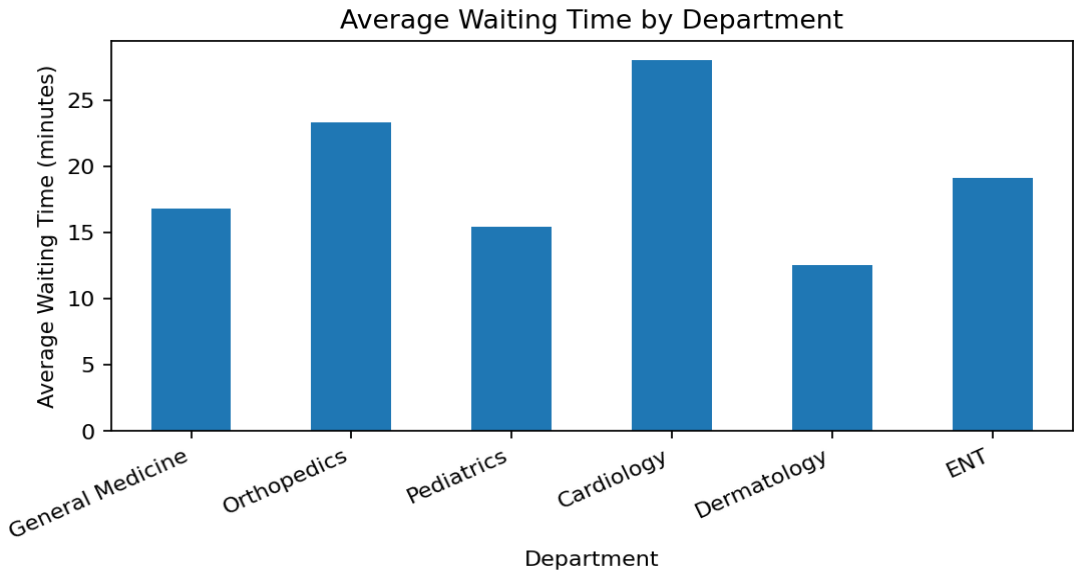
General Medicine has the highest visit count (142 records) in this generated sample.

10.2 Distribution of Waiting Time



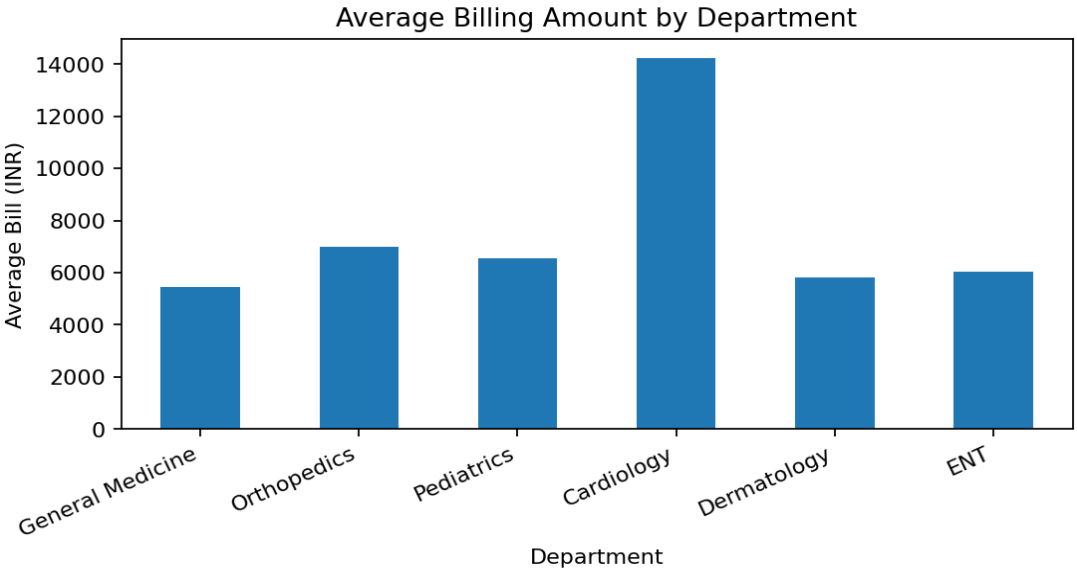
The average waiting time is 18.8 minutes. The histogram shows how waiting times are spread across visits.

10.3 Average Waiting Time by Department



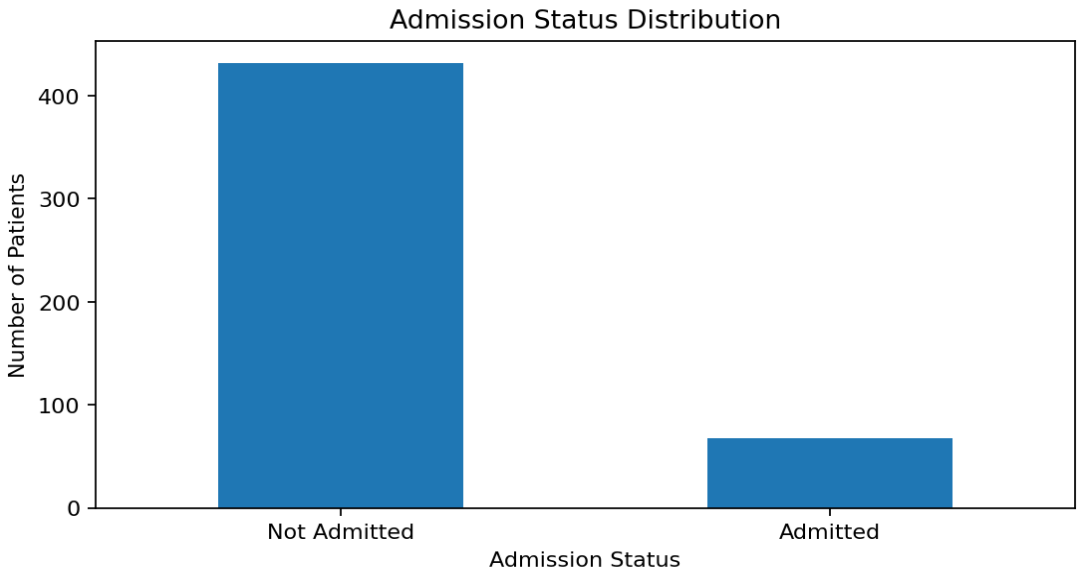
Cardiology shows the highest average wait (28.1 minutes) in this dataset.

10.4 Average Billing by Department



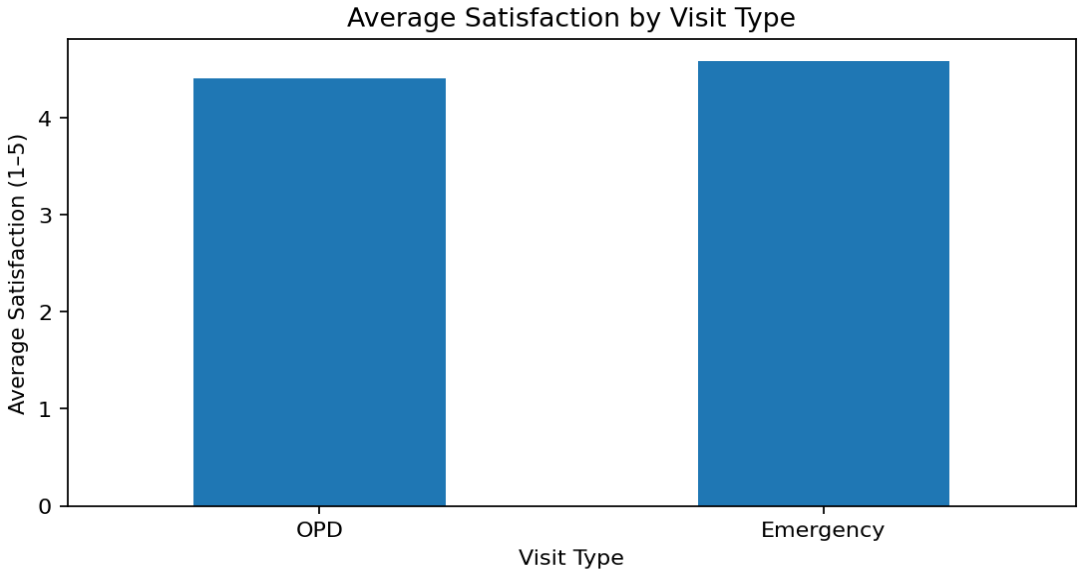
Cardiology has the highest average simulated bill (about ■14,253).

10.5 Admission Status



68 out of 500 records are labelled Admitted (13.6%).

10.6 Satisfaction by Visit Type



Average satisfaction is 4.40/5 for OPD and 4.59/5 for Emergency visits.

11. Key Findings

Department demand: General Medicine records the largest number of visits in the generated sample.

Waiting time: The highest average department wait is recorded for Cardiology. This can be used as a prompt to investigate appointment scheduling and staffing in a real-world study.

Billing: Cardiology has the highest average simulated bill. Department mix and length of stay influence the generated billing values.

Admissions: 13.6% of records have an admitted status in this sample.

Patient satisfaction: Average satisfaction differs by visit type; the values should be treated as a pattern in generated data, not as evidence about actual patient experience.

12. Recommendations (For a Real Hospital Study)

Monitor waiting time by department and time of day rather than relying only on a hospital-wide average.

Review appointment scheduling and staff allocation in departments with longer waiting times.

Study billing components separately (consultation, tests, medicines and stay charges) before drawing financial conclusions.

Collect patient feedback consistently and investigate low satisfaction scores.

Use privacy-safe, verified operational data and consult hospital professionals before making policy changes.

13. Limitations

All 500 records are artificially generated for classroom practice. They are not real patient records and do not represent any named hospital, city or population.

The generated relationships are simplified. Real hospital operations are affected by many factors that are not included here, such as time of day, staffing levels, case severity, bed availability, diagnostic tests and local pricing.

The project is descriptive and does not establish cause and effect. It is not a clinical study and cannot support diagnosis, treatment or medical decisions.

14. Conclusion

This project demonstrates how Python can be used to examine a hospital-style operations dataset. It covers data loading, quality checks, median-based cleaning, descriptive statistics, department comparisons and six visualizations.

The analysis provides a simple example of how data can help organize questions about patient flow and service performance. Any real operational decision would require accurate hospital data, privacy protection and validation by qualified staff.

15. Appendix – Dataset and Project Files

The complete 500-row dataset is supplied separately in both CSV and Excel formats so it can be opened, filtered and reused. The first ten rows are shown below as a preview; the attached spreadsheet contains all records.

Files included: Hospital_Operations_Dataset.csv, Hospital_Operations_Dataset.xlsx and Hospital_Operations_Analysis.ipynb.

Patient_ID	Age	Department	Visit_Type	Wait_Time_Minutes	Admission_Status	Satisfaction_Score
PT1001	63	Dermatology	Emergency	12	Not Admitted	5
PT1002	41	Orthopedics	OPD	32	Not Admitted	2
PT1003	49	Dermatology	OPD	15	Not Admitted	5
PT1004	38	Cardiology	Emergency	22	Admitted	5

Patient_ID	Age	Department	Visit_Type	Wait_Time_Minutes	Admission_Status	Satisfaction_Score
PT1005	36	General Medicine	OPD	21	Not Admitted	5
PT1006	26	ENT	OPD	2	Not Admitted	4
PT1007	46	Dermatology	OPD	16	Not Admitted	4
PT1008	25	Dermatology	OPD	13	Not Admitted	4
PT1009	50	General Medicine	Emergency	10	Not Admitted	5
PT1010	57	Pediatrics	OPD	21	Not Admitted	5

Academic integrity note: The dataset is synthetic and has been created only to demonstrate data-analysis techniques. It must not be presented as genuine hospital evidence.